Causal Knowledge Graphs Constructed by Circuit Propagation over Ternary Address Space

A self-contained account of the grid–graph correspondence, its backward-completion cost, and the panel that probes it

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Abstract

A causal knowledge graph (CKG) over a biochemical network is normally built by asserting edges: a curator, a text-mining pipeline, or a statistical test decides that species u influences species v, and the assertion is stored. We show that for networks admitting a steady-state circuit description, the graph need not be asserted at all. It is the fixed point of a propagation operator that the circuit already defines, and its edges are recovered rather than recorded.

The construction has three parts, each developed here from first principles with no reliance on prior work by the author. First, a metabolic network at steady state is a resistive circuit: chemical potentials $\mu_i = \mu_i^\circ + RT \ln c_i$ are node voltages, conductances $G_{ij} = k_{ij} c_i / RT$ are edge admittances, and fluxes $J_{ij} = G_{ij}(\mu_i - \mu_j)$ obey Kirchhoff's current law exactly when mass is conserved. Second, each species is assigned a point $\mathbf{S} \in [0,1]^3$ computed from the solved circuit, and that point is digitised into a base-3 address by interleaving trits across the three coordinates. Addresses are not stored in a tree: ancestry is a prefix operation on the address string, so navigation to a node's causal antecedents costs $\lceil \log_3 N \rceil$ symbol operations with no traversal and no index. Third, the propagation operator $\mathcal{T}(v_0, I, \Pi)$ that carries an initial marking through the circuit is shown to be direction-agnostic, and the CKG is exactly $\mathcal{T}$ evaluated at the trivial process: $\mathrm{cell}(v) = \mathcal{T}(v, \{v\}, \Pi_{\mathrm{rest}})$. The circuit and the graph are one map at two arguments.

We prove four structural results. Theorem 4.5 (path opacity) shows that endpoint-computed invariants cannot distinguish interior routes, so a CKG edge names an equivalence class of mechanisms and not a mechanism. Theorem 5.2 shows that backward completion restricted to physically realisable intermediate states costs $\Theta(N)$, while admitting *virtual* sub-states — triples averaging to a physical point but individually outside $[0,1]^3$ — collapses this to $\Theta(\log_3 N)$; the virtual states are the entire speedup and not an optimisation. Theorem 5.4 shows the same device is unavailable in the forward direction, which is why construction and interrogation have different costs. Theorem 7.2 gives the panel requirement: a single probe cannot separate states that collide in its own observable, and $m \geq 3$ probes with pairwise-disjoint observables are necessary for a network whose collision graph has chromatic number 3.

We then specify the interrogation layer. Reasoners are agents with disjoint internal state that never exchange content, only addresses; allocation across them is a water-filling problem with a single shadow price, so all reasoners are graded rather than selected; agreement is measured by a Kuramoto order parameter with a computable locking threshold. A drop in that order parameter localises to the newly admitted reasoner's frequency mismatch rather than to decay of the existing coordination, which we establish as theorem 7.14 and which makes the statistic diagnostic rather than merely descriptive.

Every quantitative claim is stated with the measurement that supports it or is explicitly marked as unverified. Two claims we do not make: that this construction is more accurate

than curation, and that the recovered graph is correct where the underlying kinetic parameters are wrong.

Contents

Part I
The Circuit

1 Introduction

1.1 What is being claimed

A causal knowledge graph over a biochemical system is a directed graph whose vertices are molecular species and whose edges assert influence. In the dominant construction, edges are *assertions*: they enter the graph because a curator read a paper [Caspi et al., 2020, Kanehisa and Goto, 2000], because a text-mining system extracted a relation [Percha and Altman, 2018], or because a statistical procedure rejected a null hypothesis of independence [Spirtes et al., 2000, Pearl, 2009]. The graph is then a database of claims, and its quality is the quality of its claims.

We describe a different construction. Suppose the network in question admits a steady-state description as a resistive circuit — a condition we make precise in section 2 and which holds for the metabolic networks we treat. Then there is a propagation operator, definable from the circuit alone, whose value at the trivial argument *is* the causal graph. No edge is asserted. Edges are computed, and recomputing them from the same circuit returns them again.

The claim is therefore not that this graph is better than a curated one. It is that the graph and the circuit are not two objects. Anything true of one is true of the other under a translation we exhibit, and the practical consequences — what a query costs, what a probe can and cannot resolve, what a disagreement between two derivations means — follow from the circuit's structure rather than from bookkeeping about provenance.

1.2 Why the cost structure is the interesting part

If the graph is a computation rather than a store, its useful properties are computational. Three matter here.

(i) **Backward navigation is cheap and forward construction is not.** Finding the causal antecedents of a node is a prefix operation on its address, costing $\lceil \log_3 N \rceil$ symbol comparisons. Constructing the node's address in the first place requires solving the circuit, which is $\Theta(N)$ in the number of species. The asymmetry is not an artefact of implementation; it is theorems 5.2 and 5.4, and it arises because a device available in one direction (virtual sub-states) is provably unavailable in the other.

(ii) **Storage is $O(1)$ in the number of addressable objects.** The resident state of the scheme is a coordinate table and an encoding rule. The set of objects whose address is computable is countably infinite. This is bought by recomputation, not by compression: every query pays arithmetic that a materialised index would not. Section 6 states this honestly, including the regime where an index is simply better.

(iii) **Two derivations that reach the same endpoint cannot contradict each other.** By theorem 4.5, the interior of a propagation is not recoverable from its endpoints, so it was never part of the answer. When two routes to the same query return different results, the difference measures how much of the network each route resolved, not which is true. This reclassifies a family of findings that would otherwise be read as defects; the companion paper [Sachikonye, 2026] works one such case in full.

1.3 Standing conventions

Throughout, N denotes the number of molecular species in the network under consideration, $R = 8.314 \times 10^{-3}\,\mathrm{kJ\,mol^{-1}\,K^{-1}}$ the gas constant, and T absolute temperature, with $RT = 2.577\,\mathrm{kJ\,mol^{-1}}$ at $T = 310\,\mathrm{K}$ used in all worked figures. Concentrations c_i are strictly positive. We write $\Sigma_3 = \{0, 1, 2\}$ for the ternary alphabet and Σ_3^k for words of length k. All logarithms written $\log_3$ are to base three; ln is natural.

This paper is self-contained. Every definition it uses is stated in it. Where a result is standard we cite the standard source; where a result is ours we prove it here rather than referring elsewhere.

2 The steady-state circuit

2.1 Species, reactions, and the potential

Definition 2.1 (Reaction network)**.** A *reaction network* is a pair $(\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, \ldots, N\}$ indexes molecular species and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ indexes directed elementary conversions. Each species i carries a standard chemical potential $\mu_i^\circ \in \mathbb{R}$ and a concentration $c_i > 0$. Each edge $(i, j) \in \mathcal{E}$ carries a catalytic rate constant $k_{ij} > 0$.

The physical chemistry is one equation [Atkins et al., 2018].

Definition 2.2 (Chemical potential)**.** The chemical potential of species i is

$$\mu_i \;=\; \mu_i^\circ + RT \ln c_i. \tag{1}$$

Remark 2.3. Equation (1) is the ideal-solution form. It is exact for dilute species and carries an activity-coefficient correction otherwise. We use the ideal form throughout and note in section 11 exactly what this costs.

2.2 The electrical analogy, stated as a theorem and not a metaphor

The analogy between chemical networks and electrical circuits is old [Oster et al., 1971, Peusner, 1986]; what we need is the precise statement that makes it exact rather than suggestive.

Definition 2.4 (Conductance and flux)**.** For $(i, j) \in \mathcal{E}$ define the *conductance*

$$G_{ij} \;=\; \frac{k_{ij}\, c_i}{RT} \tag{2}$$

and the *flux*

$$J_{ij} \;=\; G_{ij}\,(\mu_i - \mu_j). \tag{3}$$

Equation (3) is Ohm's law with μ as potential, J as current, and G as conductance. The content is in what the two Kirchhoff laws become.

Theorem 2.5 (Kirchhoff correspondence)**.** *Let a reaction network be at steady state. Then:*

(i) (Current law) *For every species i, $\sum_{j:(i,j)\in\mathcal{E}} J_{ij} - \sum_{j:(j,i)\in\mathcal{E}} J_{ji} = 0$ if and only if the concentration of i is stationary, i.e. mass is conserved at i.*

(ii) (Voltage law) *For every directed cycle $i_1 \to i_2 \to \cdots \to i_m \to i_1$ in $\mathcal{E}$, the potential differences sum to zero: $\sum_{r=1}^{m}(\mu_{i_r} - \mu_{i_{r+1}}) = 0$ with $i_{m+1} = i_1$. This is equivalent to the Wegscheider condition on the rate constants around the cycle.*

Proof. (i) The net rate of change of c_i is the sum of incoming minus outgoing conversion rates. Under eq. (3) the conversion rate along (i, j) is J_{ij} by construction, so stationarity of c_i is precisely the vanishing of the signed sum, which is Kirchhoff's current law at node i.

(ii) The sum telescopes: $\sum_r(\mu_{i_r} - \mu_{i_{r+1}}) = \mu_{i_1} - \mu_{i_1} = 0$ identically, because μ is a function on vertices. That the sum vanishes is therefore automatic once μ is well defined, and the substantive content is the converse direction: a set of rate constants admits a consistent potential function μ if and only if the products of forward rate constants around every cycle equal the products of reverse constants, which is the Wegscheider condition [Wegscheider, 1901, Schuster and Schuster, 1989]. A network violating it has no μ and hence no circuit representation. □

Remark 2.6 (What theorem 2.5 does and does not license)*.* It licenses treating the solved network as a resistive circuit whose node voltages are μ and whose branch currents are J. It does not license treating the circuit as a *dynamical* model: the construction is at steady state throughout, and everything downstream inherits that restriction. In particular the graph we build is time-invariant, which is the source of both its tractability and its main limitation (section 11).

Assumption 2.7 (Standing)**.** All networks considered satisfy the Wegscheider condition and are at steady state with all $c_i > 0$ and all $k_{ij} > 0$.

2.3 From the solved circuit to a point in the unit cube

The circuit solution assigns to every species three scalars. We normalise them into a bounded coordinate.

Definition 2.8 (Node coordinate)**.** For species i let $\mathcal{N}(i)$ denote the set of edges incident to i in either orientation. Define

$$\Phi_i = \textstyle\sum_{e \in \mathcal{N}(i)} |J_e|, \qquad\qquad \Gamma_i = \textstyle\sum_{e \in \mathcal{N}(i)} G_e, \tag{4}$$

and set

$$S_{\mathrm{k}}^{(i)} = \frac{\Phi_i}{\max_j \Phi_j}, \qquad S_{\mathrm{t}}^{(i)} = \frac{\Gamma_i}{\max_j \Gamma_j}, \qquad S_{\mathrm{e}}^{(i)} = \frac{\mu_i - \min_j \mu_j}{\max_j \mu_j - \min_j \mu_j}. \tag{5}$$

The *node coordinate* is $\mathbf{S}^{(i)} = (S_{\mathrm{k}}^{(i)}, S_{\mathrm{t}}^{(i)}, S_{\mathrm{e}}^{(i)}) \in [0,1]^3 =: \mathcal{S}$. Degenerate denominators are replaced by 1.

The three axes are, in order, a throughput measure, a capacity measure, and a potential measure. Their normalisation is by network extremum, which makes $\mathbf{S}$ a *relative* coordinate: it describes a species' standing within its network and is not comparable across networks without renormalisation. We flag this because it is the most common way to misuse the construction.

Definition 2.9 (Contact cost)**.** For $(i, j) \in \mathcal{E}$ the *contact cost* is the Euclidean distance

$$d(i, j) \;=\; \big\|\mathbf{S}^{(i)} - \mathbf{S}^{(j)}\big\|_2. \tag{6}$$

The *contact map* of the network is the assignment $(i, j) \mapsto d(i, j)$ over $\mathcal{E}$.

Proposition 2.10 (Contact map is a function of the circuit alone)**.** *Under theorem 2.7, the contact map is uniquely determined by $(\{\mu_i^\circ\}, \{c_i\}, \{k_{ij}\}, \mathcal{E}, T)$. No further input enters.*

Proof. μ is determined by eq. (1); G by eq. (2); J by eq. (3); the three normalisers in eq. (5) are network extrema over those quantities; d is a function of the resulting coordinates by eq. (6). Each step is a total function of its predecessors. □

Theorem 2.10 is the formal content of "the graph is not asserted". Everything appearing in the contact map was already present in the kinetic description; the map introduces no new facts, only a different presentation of the same ones.

2.4 Reference computation

We fix a reference implementation so that later numerical claims are checkable. The reference computes eqs. (1) to (6) over four networks — glycolysis, the citric acid cycle, oxidative phosphorylation, and the EGFR/MAPK cascade — with literature kinetic parameters, and runs eighteen consistency checks over the result. The checks cover: agreement of the flux sum with the current law at every internal node; invariance of the contact map under relabelling of species; irreducibility of the coordinate triple (no axis is a function of the other two on the tested networks); and the rank-correlation statistic of theorem 2.11 below. All eighteen pass, and all reported numbers other than timestamps are reproduced bit-identically across runs, which establishes determinism of the pipeline but of course not its correctness against biology.

Definition 2.11 (Triple coherence)**.** Let $\rho(\cdot,\cdot)$ denote Spearman rank correlation [Spearman, 1904] across species. The *triple coherence* of a network is

$$\mathcal{R} \;=\; \tfrac{1}{3}\big[\rho(S_{\mathrm{k}},S_{\mathrm{t}})+\rho(S_{\mathrm{t}},S_{\mathrm{e}})+\rho(S_{\mathrm{k}},S_{\mathrm{e}})\big]. \tag{7}$$

Coherence near 1 indicates the three axes are ranking species similarly and the coordinate is effectively one-dimensional; coherence near 0 indicates they carry independent information. It is a diagnostic on the network, not a quality score: a network with $\mathcal{R}\approx 1$ is one where the address gains little from being three-dimensional.

Part II
The Address Space

3 Ternary addressing

3.1 Why base three

The coordinate $\mathbf{S}$ lives in a three-dimensional cube. A digitisation that refines all three axes in lockstep, one symbol per axis per refinement, has a natural radix equal to the number of axes. This is a statement about the data, not the hardware: the implementation is binary throughout, and the ternary structure is in the encoding [Knuth, 1997].

The information content of one ternary symbol (a *trit*) is $\log_2 3 \approx 1.585$ bits. A binary encoding of the same three-way choice needs $\lceil \log_2 3\rceil = 2$ bits, wasting 0.415 bits per refinement, or about 26%. That overhead is secondary. The primary reason is structural: with two bits per step the map from bit position to coordinate axis requires an extra interpretive layer, whereas base three makes axis identity positional.

3.2 The encoding

Definition 3.1 (Trit expansion)**.** For $x \in [0,1)$ the *trit expansion* is the sequence $\tau_1(x), \tau_2(x), \dots \in \Sigma_3$ with $\tau_r(x) = \lfloor 3^r x\rfloor \bmod 3$, so that $x = \sum_{r\ge 1}\tau_r(x)\,3^{-r}$. For $x = 1$ set $\tau_r(1) = 2$ for all r.

Definition 3.2 (Interleaved address)**.** Fix a depth $m \in \mathbb{N}$ and put $k = 3m$. The *address* of a node with coordinate $\mathbf{S} = (S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}})$ is the word

$$\mathrm{addr}_k(\mathbf{S}) \;=\; \big(\tau_1(S_{\mathrm{k}}),\tau_1(S_{\mathrm{t}}),\tau_1(S_{\mathrm{e}}),\; \tau_2(S_{\mathrm{k}}),\tau_2(S_{\mathrm{t}}),\tau_2(S_{\mathrm{e}}),\; \dots,\; \tau_m(S_{\mathrm{k}}),\tau_m(S_{\mathrm{t}}),\tau_m(S_{\mathrm{e}})\big) \in \Sigma_3^k. \tag{8}$$

The *encoding rule* is the triple (radix 3, interleave order $(S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}})$, domain $[0,1]$).

Interleaving rather than concatenating is essential. Under interleaving, a shared prefix of length $3r$ means the two nodes agree to resolution 3^{-r} on *all three* axes simultaneously; under concatenation a shared prefix would mean agreement on the first axis only, and prefix length would carry no information about the other two.

Definition 3.3 (Scheme and cache)**.** The *scheme* Σ is the pair (coordinate construction of theorem 2.8, encoding rule of theorem 3.2). A *cache* K is any materialised set of computed addresses. A query is *cache-free* if its output is a function of Σ and the query alone.

Proposition 3.4 (Resident state is $O(1)$ in addressable objects)**.** $|\Sigma|$ *is a fixed finite quantity independent of the number of objects whose address may be requested, while the set of addressable coordinates is the whole of* $\mathcal{S}$ *and hence uncountable (its image under* addr_k *being* Σ_3^k*, of size* 3^k*).*

Proof. Σ consists of the formulae eqs. (1) to (5) and the three constants of the encoding rule. None of these is indexed by an object. Given any $\mathbf{S} \in \mathcal{S}$, eq. (8) returns a word without consulting any stored record. Hence no object need have been seen for its address to be computable, and $|\Sigma|$ does not grow when objects are added. □

Remark 3.5 (The honest form of theorem 3.4)*.* Storage is $O(1)$ where an index is $O(n)$, and this is bought by *recomputation*, not by cleverness about compression. Every query pays the arithmetic of eqs. (1) to (8) that it would not pay against a materialised index. For a corpus that is fixed, small, and queried often, the index is simply better and nothing here argues otherwise. The regime in which the cache-free scheme wins is the one where the query set is not known in advance — which, for a network under active perturbation or extension, is the usual regime.

3.3 Ancestry is a prefix operation

Here is the point at which the construction departs most sharply from the data-structure reading of the word "trie".

Definition 3.6 (Address trie, nominal)**.** The *address trie* of depth k is the complete ternary tree whose 3^k leaves are the words of Σ_3^k and whose internal vertices are the proper prefixes, with the empty word at the root.

Proposition 3.7 (The trie is never built)**.** *Let* $a \in \Sigma_3^k$ *be an address. Then:*

(i) the parent of a *in the trie of theorem 3.6 is* $a_{1:k-1}$*, obtained by deleting the last symbol;*

(ii) the number of steps from a *to the root is exactly* $|a| = k$*;*

(iii) the nearest common ancestor of a *and* b *is their longest common prefix.*

Each is computable from the address string alone, in time linear in k*, with no tree materialised, no pointer dereferenced, and no search performed.*

Proof. Immediate from theorem 3.6: the vertex set is the prefix-closed set of words, the parent relation is prefix extension by one symbol, and depth is word length. The stated operations are string operations on a. □

Remark 3.8*.* Theorem 3.7 is why we say the trie is an *addressing scheme* rather than a data structure. Statements of the form "the query costs $\log_3 N$" are, in this construction, statements that the address is $\log_3 N$ symbols long — not that a balanced tree of N items was descended. The distinction matters because the second reading invites the objection "you still had to build the tree", and there is no tree.

A direct check of theorem 3.7(ii) over depths 2 through 10 (so $N = 3^d$ from 9 to 59 049), sampling up to 100 random addresses at each depth, returns step counts whose minimum, maximum, and expected value coincide at every depth: zero variance, exact agreement. This is not evidence for a bound; it is a consistency check that the implementation computes $|a|$ and nothing else.

3.4 Similarity as shared prefix

Definition 3.9 (Longest common prefix and address similarity)**.** For $a, b \in \Sigma_3^k$ let $\ell(a,b) = \max\{r : a_{1:r} = b_{1:r}\}$. For composite objects u, v presented as n-tuples of addressable parts, the *address similarity* is

$$\sigma(u,v) \;=\; \frac{1}{nk}\sum_{i=1}^{n} \ell(\mathrm{addr}_k(u_i), \mathrm{addr}_k(v_i)). \tag{9}$$

Proposition 3.10 (Prefix similarity is a partition statement)**.** *$\ell(a,b) \geq 3r$ if and only if a and b lie in the same depth-r subcube of $\mathcal{S}$ under theorem 3.2, i.e. their coordinates agree to within 3^{-r} on every axis after truncation.*

Proof. By theorem 3.2 the first $3r$ symbols of an address are exactly $\tau_1, \ldots, \tau_r$ of each of the three coordinates. Equality of those symbols is equality of the truncated base-3 expansions to r places on each axis, which is precisely co-membership in a common subcube of side 3^{-r}. □

Remark 3.11 (What σ is not)*.* σ is not a proxy for the Euclidean contact cost of theorem 2.9. The address is lexicographic, so two nodes differing in the *first* trit of S_{k} share no prefix at all however close they are in coordinate distance; the induced separation $1-\sigma$ is a coarsened and discontinuous function of Euclidean distance by construction. The two quantities do different jobs: d measures how far apart two species sit in the solved circuit, σ measures whether they fall in the same cell of the address partition. Neither substitutes for the other, and we use both, never interchangeably.

Remark 3.12 (Empirical calibration and its ceiling)*.* Against an external physicochemical reference distance built from three scales not used anywhere in the scheme, address separation attains a rank correlation of $\rho = 0.382$ on an exhaustive set of 190 pairs, while 200 random permutations of the coordinate assignment give mean $\rho = -0.004$ with maximum 0.252, so no permutation reaches the observed value ($p < 0.005$ empirically). The correct reading is narrow: the coordinate assignment carries information that no relabelling reproduces. It is not that σ is a good metric for physicochemical distance. It is not, for the reason in theorem 3.11, and we decline to describe $\rho = 0.382$ as more than moderate.

3.5 Choosing the depth

The depth k is the only free parameter of the encoding. It is fixed by an entropy-margin argument rather than by taste.

Proposition 3.13 (Capacity condition)**.** *To distinguish the members of a classification tier carrying Shannon entropy H bits, a depth-k address suffices only if*

$$k\log_2 3 \;\geq\; H. \tag{10}$$

Proof. A depth-k address takes 3^k values and hence carries at most $\log_2 3^k = k\log_2 3$ bits. A code distinguishing outcomes of entropy H requires at least H bits by the source-coding theorem [Shannon, 1948, Cover and Thomas, 2006]. □

Example 3.14 (Tier selection). For a tier with M roughly equiprobable members, $H \approx \log_2 M$. Table 1 records the resulting minimum depths for three nested tiers of a protein superfamily and for the reference glycolysis network.

Table 1: Depth selection by eq. (10). Margin is $k \log_2 3 - H$ in bits.

Tier	members	H (bits)	k	margin
Family	18	4.17	3	+0.58
Isoform	57	5.83	6	+3.68
Allele	> 300	8.23	9	+6.04
Glycolysis (reference)	≈ 10	3.32	3	+1.43

Proposition 3.15 (Partial-address recovery)**.** *Suppose a fraction $\theta \in (0, 1]$ of a depth-k address is observed and the remainder is unconstrained, and model the residual ambiguity as uniform over the unobserved symbols. Then the probability of a unique correct completion against a tier of entropy H is bounded below by*

$$P_{\text{correct}} \;\geq\; 1 - \exp\bigl(-3(\theta k \log_2 3 - H)\bigr) \tag{11}$$

whenever the bracket is positive.

Proof sketch. The observed prefix restricts the candidate set to a subcube containing an expected $2^{H-\theta k \log_2 3}$ tier members. Treating collisions as a Poisson process in the number of surplus bits gives a collision probability $\exp(-\lambda)$ with λ proportional to the surplus; the constant 3 is the radix-induced branching per symbol position, and the bound follows. The estimate is an approximation under a uniformity assumption on tier placement and is stated as a bound rather than an identity for that reason. □

At the isoform tier ($k = 6$, $H = 5.83$) with $\theta = 0.7$, eq. (11) gives $P_{\text{correct}} \approx 0.92$. The practical reading: a node can be identified from an incomplete circuit solve, and the incompleteness one can tolerate is quantified rather than hoped for.

Part III
Propagation

4 The propagation operator

4.1 Contact graph and admissible sets

Definition 4.1 (Contact graph)**.** The *contact graph* of a network is the weighted undirected graph on $\mathcal{V} \cup \{\mathfrak{m}\}$ whose edges are $\mathcal{E}$ with weights $d(i, j)$ from eq. (6), together with an edge from every species to a distinguished *medium* vertex $\mathfrak{m}$ of weight β.

The medium vertex represents the solvent-mediated contact every species has with every other. It is what prevents the graph from decomposing into disconnected components when the reaction edges alone would do so, and it carries the following floor.

Assumption 4.2 (Contact floor)**.** There is a constant $\beta > 0$ such that every contact in the contact graph has weight at least β.

Theorem 4.2 is the single physical fact the propagation results rest on. It says no contact is free: any interaction between two species, including through the medium, costs something bounded away from zero. Its consequences are disproportionate to its modesty, as the next results show.

Definition 4.3 (Process and propagation)**.** A *process* Π is a specification of which contacts are active. Given a starting vertex v_0, an initial marked set $I \subseteq \mathcal{V}$, and a process Π, the *propagation operator*

$$\mathcal{T}(v_0, I, \Pi) \;=\; \mathcal{A}(v_0, x^*, \Pi) \tag{12}$$

returns the *admissible set*: the set of vertices reachable from v_0 under Π by a sequence of contacts whose accumulated weight converges to the fixed point x^*.

Theorem 4.4 (Inadmissibility has one source)**.** *Under theorem 4.2, a vertex v fails to lie in $\mathcal{T}(v_0, I, \Pi)$ if and only if no contact sequence from v_0 to v under Π converges. In particular, local misalignment between two adjacent vertices never by itself causes inadmissibility.*

Proof. ($\Leftarrow$) If no sequence converges, v is not in the admissible set by theorem 4.3.

($\Rightarrow$) Suppose some sequence from v_0 to v converges to x^*. By theorem 4.2 every contact in that sequence has weight $\geq \beta > 0$, so the sequence is finite: a convergent sum of terms each at least β has at most x^*/β terms. A finite converging sequence witnesses membership, so $v \in \mathcal{A}(v_0, x^*, \Pi)$. Hence failure of membership implies no sequence converges. Local misalignment at a single contact cannot produce failure, since it can at worst increase that contact's weight, and increased weight shortens rather than destroys the admissible sequence bound. □

Theorem 4.4 is doing real work later: it is the reason two derivations reaching the same fixed point cannot be in contradiction, because the only way to fail is global non-convergence, which is a property of the pair (v_0, v) and not of any route between them.

4.2 Path opacity

Theorem 4.5 (Path opacity)**.** *Let P and Q be two distinct contact sequences from v_0 to v, both converging to the same fixed point x^* under the same process Π. Then $\mathcal{T}$ assigns them the same value, and no invariant computed from the endpoints (v_0, v, x^*) distinguishes P from Q.*

Proof. By eq. (12), $\mathcal{T}$ is a function of (v_0, I, Π) and returns $\mathcal{A}(v_0, x^*, \Pi)$, a set defined by existence of a converging sequence, not by choice of one. Both P and Q witness the same membership statements, so the returned sets coincide. Any invariant that is by hypothesis a function of (v_0, v, x^*) is then constant on $\{P, Q\}$, since those three arguments agree. □

Corollary 4.6 (Output is a class, not a route)**.** *The value of $\mathcal{T}$ names an equivalence class of contact sequences under the relation "same endpoints, same fixed point". Any two members of the class are interchangeable in every downstream computation that consumes only $\mathcal{T}$'s value.*

Remark 4.7 (This is a limitation stated as a theorem). Theorem 4.5 says the construction *cannot* tell you the mechanism. An edge of the recovered CKG asserts that influence propagates, and refuses to say by which route. Where the route is the object of interest — as in mechanistic enzymology — this construction is the wrong tool, and no amount of depth k repairs that, because the deficiency is in what $\mathcal{T}$ is a function of. We return to the consequences in section 11.

4.3 Direction agnosticism, and the CKG as a value of $\mathcal{T}$

Theorem 4.8 (The table is direction-agnostic)**.** *Let $\mathcal{A}^{\rightarrow}(v_0, x^*, \Pi)$ be the admissible set computed by extending contact sequences forward from v_0, and $\mathcal{A}^{\leftarrow}(v_0, x^*, \Pi)$ the set computed by extending them backward from the targets. Under theorem 4.2 the two coincide.*

Proof. Membership of v in either set is, by theorem 4.3, the existence of a converging contact sequence between v_0 and v under Π. The contact graph of theorem 4.1 is undirected and its weights are symmetric, so a sequence witnessing membership read forward is a sequence witnessing membership read backward, with the same accumulated weight and hence the same fixed point. Existence is therefore direction-independent, and the two sets have the same members. □

Remark 4.9 (What theorem 4.8 costs and what it does not). It says forward and backward interrogation return the *same answer.* It does not say they cost the same, and theorems 5.2 and 5.4 show they do not. This separation — same value, different price — is what makes a two-directional probe worth building: the two directions are not adjudicating a disagreement about content, they are trading cost against availability.

Theorem 4.10 (The CKG is the propagation at rest)**.** *Define the* resting process Π_{rest} *to be the process in which all contacts of the solved circuit are active and no external drive is applied. Then for every species v,*

$$\text{cell}(v) \;=\; \mathcal{T}(v, \{v\}, \Pi_{\text{rest}}), \tag{13}$$

where $\text{cell}(v)$ *is the set of species causally reachable from v in the recovered causal knowledge graph. Consequently the CKG is the propagation operator evaluated at the trivial marking, and the propagation is the CKG carried forward under a non-trivial process.*

Proof. Take the definition of a CKG edge to be: $v \to w$ whenever a perturbation at v changes the steady state at w under Π_{rest}. By theorem 2.5, a perturbation at v propagates through the circuit along contacts, and w responds if and only if some contact sequence from v to w carries the change, i.e. converges. By theorem 4.3 that is exactly membership of w in $\mathcal{T}(v, \{v\}, \Pi_{\text{rest}})$. Transitive closure of the edge relation is the reachable set, giving eq. (13). The converse statement follows by substituting a general Π for Π_{rest} in the same argument. □

Theorem 4.10 is the paper's central identification and the reason its title pairs "circuit" with "CKG". They are one map at two arguments: at the resting argument it is a graph, at a driven argument it is a propagation. There is no translation step between them to get wrong, and no synchronisation problem between a stored graph and a live model, because there is no stored graph.

5 The cost of going backward

5.1 Virtual sub-states

Backward completion asks: given a node's coordinate, reconstruct a sequence of antecedent coordinates consistent with it. The natural constraint is that every reconstructed intermediate be physically realisable, i.e. lie in $\mathcal{S} = [0, 1]^3$. That constraint is exactly what makes the problem expensive.

Definition 5.1 (Sub-state decomposition)**.** A *sub-state decomposition* of $s \in \mathcal{S}$ of order p is a tuple $(s_1, \ldots, s_p) \in (\mathbb{R}^3)^p$ with

$$\frac{1}{p}\sum_{r=1}^{p} s_r \;=\; s. \tag{14}$$

The decomposition is *physical* if every $s_r \in \mathcal{S}$, and *virtual* if some $s_r \notin \mathcal{S}$.

Virtual sub-states are not a modelling fiction to be apologised for. They are the arithmetic of the reconstruction: nothing in eq. (14) requires the summands to be individually realisable, and forbidding them is an extra constraint with a measurable price.

Theorem 5.2 (Collapse)**.** *Consider backward completion of a depth-k address over a network of $N = 3^k$ addressable cells.*

(i) Restricted to physical decompositions, the completion requires $\Theta(N)$ operations in the worst case.

(ii) Permitting virtual decompositions, the completion requires $\Theta(\log_3 N) = \Theta(k)$ operations.

Proof. (ii) With virtual decompositions permitted, the antecedent of an address a is its prefix $a_{1:|a|-1}$ (theorem 3.7(i)), because the coordinate of the parent cell is the cell-average of its three children and eq. (14) with $p = 3$ is satisfied by the children's coordinates whether or not each lies in $\mathcal{S}$ after the averaging is inverted. Iterating to the root takes $|a| = k$ deletions, each $O(1)$, giving $\Theta(k) = \Theta(\log_3 N)$.

(i) Restricted to physical decompositions, the parent's coordinate is constrained to admit three children all in $\mathcal{S}$. Near the boundary of $\mathcal{S}$ this fails: for a cell whose coordinate lies within 3^{-k} of a face, some of the three summands required by eq. (14) must have a negative component or a component exceeding 1. The completion must then search for an alternative antecedent elsewhere in the cube. In the worst case the set of cells admitting a fully physical decomposition is a vanishing fraction of the N cells, and locating one requires examining $\Theta(N)$ of them, since no prefix relation orders the search — physicality is not a prefix-closed property. □

Corollary 5.3 (The speedup is the virtual states)**.** *The gap between* $\Theta(N)$ *and* $\Theta(\log_3 N)$ *in theorem 5.2 is attributable entirely to admitting decompositions with components outside* $\mathcal{S}$*. No reorganisation of the physical-only search closes it, because the obstruction is that physicality is not prefix-closed.*

A direct measurement of the virtual fraction supports the mechanism. Sampling 1000 decompositions at each of four spread parameters $M \in \{1, 2, 5, 10\}$ — M controlling how widely the summands are dispersed about their mean — gives virtual fractions of 0.917, 0.975, 0.996, and 0.998 respectively: monotone in M, and above 90% even at the tightest setting, with the mean of each decomposition recovering its target in every case. The reading is that virtual decompositions are the generic case and physical ones the exception, which is what theorem 5.2(i) predicts.

Theorem 5.4 (Forward asymmetry)**.** *Forward construction of a node's address from the circuit cannot use virtual sub-states, and costs* $\Theta(N)$*.*

Proof. Forward construction evaluates eq. (5), whose three normalisers are network extrema $\max_j \Phi_j$, $\max_j \Gamma_j$, $\min_j \mu_j$, $\max_j \mu_j$. Each extremum is a function of all N species and cannot be evaluated from a prefix. Furthermore the intermediate quantities Φ_i, Γ_i, μ_i are physical observables of the solved circuit, not free summands: there is no analogue of eq. (14) permitting them to leave their ranges, because they are read off eqs. (1) to (3) rather than posited. Hence the completion device of theorem 5.2(ii) is unavailable and the cost is that of evaluating the extrema, $\Theta(N)$. □

Remark 5.5 (Where the asymmetry comes from)*.* Backward completion is unconstrained inference over a lattice; forward construction is constrained evaluation of a physical model. Virtual sub-states are legitimate in the first because nothing there claims the intermediates are states of anything. They are illegitimate in the second because there the intermediates are exactly claims about the circuit. The asymmetry of theorems 5.2 and 5.4 is therefore not a quirk of the encoding; it reflects that reconstruction and simulation answer different questions.

Table 2: Cost summary. N is species count, $k = \lceil \log_3 N \rceil$.

Operation	Cost	Source
Solve circuit, all coordinates	$\Theta(N + \lvert\mathcal{E}\rvert)$	eqs. (1) to (5)
Address one node, coordinate known	$\Theta(k)$	theorem 3.2
Address one node, from circuit	$\Theta(N)$	theorem 5.4
Ancestors of a node	$\Theta(k)$	theorem 3.7
Backward completion, physical only	$\Theta(N)$	theorem 5.2(i)
Backward completion, virtual allowed	$\Theta(k)$	theorem 5.2(ii)
Similarity $\sigma(u, v)$, n parts	$\Theta(nk)$	theorem 3.9
Resident state	$\Theta(1)$	theorem 3.4

6 The empty-scheme property

Theorem 6.1 (Answers without entries). *Every query of the forms*

$$\mathsf{Identify}(u) = \mathrm{addr}_k(\mathbf{S}^{(u)}), \qquad \mathsf{Similar}(u,v) = \sigma(u,v), \qquad \mathsf{Predict}(u;C) = \operatorname*{argmax}_{c\in C} \sigma(u,c)$$

is cache-free in the sense of theorem 3.3.

Proof. Each output is by definition a composition of eq. (5), eq. (8), and eq. (9), all of which are total functions of Σ and their arguments. No step reads a stored record keyed by an object. □

Corollary 6.2 (Answering for the unseen). *A query may be answered for an object never previously processed, provided its coordinate is computable. An exact-match index over previously stored objects returns nothing for such an object.*

Proposition 6.3 (The cache is not load-bearing). *Let A_K and $A_\emptyset$ denote the answers to a fixed query batch with a cache materialised and deleted respectively. Then $A_K = A_\emptyset$, and $|\Sigma|$ is unchanged by either.*

Proof. By theorem 6.1 no query reads K; hence deleting K changes no input to any query, and the outputs coincide. $|\Sigma|$ is defined without reference to K by theorem 3.3. □

Remark 6.4 (A distinction that is easy to lose). A materialised set of addresses for a fixed corpus of n objects costs $O(nk)$ trits. That artefact is a *compressed dictionary*, and compression against structural representations can be dramatic. It is not an empty scheme, and the difference is not pedantry: a compressed dictionary cannot answer for an object it does not contain, and by theorem 6.2 the scheme can. Reporting the cache size as though it were the resident state conflates the two and inverts the claim.

Part IV
Interrogation

7 Why more than one probe

7.1 Collision, and the discrimination bound

Any single observable partitions states into fibres, and states within a fibre are indistinguishable to it. This is not a defect of a particular instrument; it is what an observable is.

Definition 7.1 (Probe, collision graph). A *probe* is a map $\pi : \mathcal{X} \to \mathcal{Y}_\pi$ from a state set $\mathcal{X}$ to that probe's readout set. States $x \neq x'$ *collide under* π if $\pi(x) = \pi(x')$. Given a family $\mathsf{P} = \{\pi_1, \ldots, \pi_m\}$, the *collision graph* $\mathcal{C}(\mathsf{P})$ has vertex set $\mathcal{X}$ and an edge $\{x, x'\}$ whenever x, x' collide under *every* $\pi \in \mathsf{P}$.

Theorem 7.2 (Discrimination requires disjoint observables). *A family P separates all states, i.e. $\mathcal{C}(\mathsf{P})$ has no edges, if and only if for every pair $x \neq x'$ there is some $\pi \in \mathsf{P}$ with $\pi(x) \neq \pi(x')$. If the pairwise-collision relation of the individual probes requires χ colours to cover $\mathcal{X}$'s pairs — that is, if no $\chi - 1$ probes suffice — then $|\mathsf{P}| \geq \chi$.*

Proof. The first statement is the definition of separation unfolded. For the second, assign to each unordered pair $\{x, x'\}$ the set of probes that separate it. If some $\mathsf{P}' \subseteq \mathsf{P}$ with $|\mathsf{P}'| = \chi - 1$ separated all pairs, then every pair would have a separator in P', contradicting the hypothesis that no $\chi - 1$ probes suffice. Hence $|\mathsf{P}| \geq \chi$. □

Theorem 7.2 is content-free until one exhibits a system with $\chi \geq 3$. Spectroscopy of a seven-state catalytic cycle supplies one. Under a single absorption probe, two pairs of states collide: the resting and oxygen-bound forms differ by 1 nm in band position (417 nm against 418 nm), and two high-valent intermediates differ by 3 nm (367 nm against 370 nm) — both within linewidth. A magnetic-resonance probe separates the first pair, since it reads spin state directly and the two forms differ there. A vibrational probe separates the second, since one of the two carries a metal–oxo stretch the other lacks. Neither of the two additional probes separates the pair the other handles. So no single probe suffices, and $\chi \geq 2$ on this system [Denisov et al., 2005, Rittle and Green, 2010].

It is tempting to conclude that three probes are therefore necessary, and we originally did. Exhaustive search over probe subsets shows otherwise, and the correction is instructive rather than incidental. Once the state table carries a fourth observable that is independently available on this system — formal iron oxidation state, read by Mössbauer or X-ray absorption — the minimum drops to two. Oxidation state separates *both* residual pairs at once, since the resting and oxygen-bound forms differ as Fe(III) against Fe(II) and the two high-valent intermediates as Fe(III) against Fe(IV). The absorption probe then handles everything else, and the vibrational probe becomes redundant. The minimum separating family is $\{\text{absorption}, \text{oxidation state}\}$, of size two.

Corollary 7.3 (Panel floor)**.** *For a state space $\mathcal{X}$ and an available observable set Π, the panel floor is $\chi(\mathcal{X}, \Pi) = \min\{|\mathsf{P}| : \mathsf{P} \subseteq \Pi,\ \mathcal{C}(\mathsf{P})$ edgeless$\}$, and a panel must contain at least χ members with pairwise non-substitutable observables. Redundant copies of one probe do not raise discrimination, so χ is unchanged by duplicating any member.*

Remark 7.4 (The floor is computed, not constant). χ is a joint property of the state set *and* the observable set, and must be computed per system. It is not a constant of the method, and in particular it is not three: on the seven-state cycle above it is two once oxidation state is admitted, and three if that observable is unavailable. Enlarging Π can only lower χ; enlarging $\mathcal{X}$ can only raise it. Reporting a panel size without the search that produced it asserts a floor rather than establishing one.

Remark 7.5. This is the argument for a panel of reasoners, stated in instrument terms. The members are not voters whose agreement is averaged; they are non-overlapping observables whose *disjointness* is what buys resolution. A panel of m identical reasoners has the discriminating power of one, for every m. What the correction above adds is that disjointness is the operative property and cardinality is downstream of it: a smaller panel of more separating observables beats a larger panel of overlapping ones.

7.2 Reasoners as disjoint agents

Definition 7.6 (Reasoner)**.** A *reasoner* A_i is a triple $(\Gamma_i, \omega_i, \gamma_i)$ where Γ_i is its internal state graph, ω_i its intrinsic update frequency, and $\gamma_i : [0, \infty) \to \mathbb{R}$ its *yield function*, mapping allocated effort a to expected resolution gain. Two reasoners are *disjoint* if $\Gamma_i \cap \Gamma_j = \emptyset$ as labelled graphs.

Assumption 7.7 (Yield regularity)**.** Each γ_i is continuously differentiable, strictly increasing, and strictly concave on $[0, \infty)$, with $\gamma_i(0) = 0$.

Concavity is an assumption about the *environment* the reasoner queries — diminishing returns to further probing of a fixed network — not about the reasoner's internal architecture. We flag this because it is the assumption most likely to fail: a reasoner with a threshold effect (nothing until enough effort, then a jump) violates it, and theorem 7.9 does not apply to such a panel.

Definition 7.8 (Disjointness certificate)**.** Reasoners A_i, A_j carry a *disjointness certificate* if (a) the vertex label sets of Γ_i, Γ_j are disjoint, and (b) no weighted graph isomorphism $\Gamma_i \to \Gamma_j$ exists.

Condition (b) is what makes the certificate non-vacuous: without it, two reasoners could be relabelled copies of one another and satisfy (a) trivially. A check that can only pass — for instance, verifying disjointness of reasoners that were *constructed* disjoint by fiat — tests nothing, and we treat such a check as absent rather than as passed (see theorem 10.1).

7.3 Allocation by water-filling

Given a budget B of total effort, how much goes to each reasoner? The answer is not a selection problem.

Theorem 7.9 (Water-filling allocation)**.** *Under theorem 7.7, the allocation maximising $\sum_i \gamma_i(a_i)$ subject to $\sum_i a_i = B$, $a_i \geq 0$, is characterised by a single scalar $p^\star > 0$ with*

$$\gamma_i'(a_i^\star) = p^\star \quad \text{if } a_i^\star > 0, \qquad \gamma_i'(0) \leq p^\star \quad \text{if } a_i^\star = 0. \tag{15}$$

The optimum is unique, and $p^\star$ is the unique root of $\sum_i (\gamma_i')^{-1}(p) = B$ on the range where the inverse is defined.

Proof. The objective is strictly concave and the feasible set is a compact convex simplex, so a unique maximiser exists. The Karush–Kuhn–Tucker conditions [Boyd and Vandenberghe, 2004] for the Lagrangian $L = \sum_i \gamma_i(a_i) - p(\sum_i a_i - B) + \sum_i \nu_i a_i$ give $\gamma_i'(a_i) - p + \nu_i = 0$ with $\nu_i \geq 0$ and $\nu_i a_i = 0$. For $a_i > 0$ complementary slackness forces $\nu_i = 0$ and hence $\gamma_i'(a_i) = p$; for $a_i = 0$ it gives $\gamma_i'(0) = p - \nu_i \leq p$. Setting $p^\star = p$ yields eq. (15). Strict concavity makes γ_i' strictly decreasing hence invertible, and $p \mapsto \sum_i (\gamma_i')^{-1}(p)$ is strictly decreasing, so the budget equation has a unique root, computable by bisection. □

Corollary 7.10 (No reasoner is switched off by policy)**.** *A reasoner receives zero allocation only in the boundary case $\gamma_i'(0) \leq p^\star$, i.e. only when its marginal yield at zero effort is already below the common price. Otherwise every reasoner is engaged with a positive graded allocation.*

Theorem 7.10 is the substantive difference from a selection rule such as a knapsack: there is one price, not a cutoff on identities, and a reasoner drops out because of its own marginal yield rather than because a ranking excluded it.

7.4 Agreement by phase locking

Reasoners must be said to agree on something. We take agreement to be synchronisation of their update phases on a shared address, which has the advantage of a computable threshold.

Definition 7.11 (Order parameter)**.** For phases $\phi_1, \ldots, \phi_M$ define

$$\mathcal{R}\, e^{\mathrm{i}\psi} \;=\; \frac{1}{M} \sum_{j=1}^{M} e^{\mathrm{i}\phi_j}, \qquad \mathcal{R} \in [0,1]. \tag{16}$$

Definition 7.12 (Coupled panel)**.** A panel with coupling strength K evolves by

$$\dot{\phi}_j \;=\; \omega_j + \frac{K}{M} \sum_{l=1}^{M} \sin(\phi_l - \phi_j), \qquad j = 1, \ldots, M, \tag{17}$$

which is the Kuramoto model [Kuramoto, 1975, Acebrón et al., 2005].

Theorem 7.13 (Locking threshold)**.** *Let the intrinsic frequencies ω_j be drawn from a unimodal symmetric density g with mean $\bar{\omega}$. In the large-M limit,* eq. (17) *admits a partially synchronised solution ($\mathcal{R} > 0$) if and only if $K > K_c$ with*

$$K_c \;=\; \frac{2}{\pi\, g(\bar{\omega})}. \tag{18}$$

For $K < K_c$ the incoherent state $\mathcal{R} = 0$ is the only solution.

Proof reference. This is Kuramoto's self-consistency result; the derivation sets $\mathcal{R} = K\mathcal{R}\int_{-\pi/2}^{\pi/2}\cos^2\theta\, g(\bar{\omega}+K\mathcal{R}\sin\theta)\,\mathrm{d}\theta$ and takes $\mathcal{R}\to 0^+$, giving $1 = K\pi g(\bar{\omega})/2$ [Kuramoto, 1975, Strogatz, 2000]. □

The practical content of theorem 7.13 is that the coupling needed for agreement is $O(\Delta\omega)$ in the spread of intrinsic frequencies: a panel of reasoners with similar update rates locks cheaply, and one with disparate rates does not lock at all below threshold. The threshold is computed from the observed spread rather than assumed.

Proposition 7.14 (A fall in $\mathcal{R}$ localises to the newcomer)**.** *Let a panel of M locked reasoners with $\mathcal{R}_{\text{old}}$ admit one further reasoner whose intrinsic frequency ω_{M+1} lies outside the locked band. Then the new order parameter satisfies*

$$\mathcal{R}_{\text{new}} \;\le\; \frac{M\,\mathcal{R}_{\text{old}}+1}{M+1}, \tag{19}$$

with equality only if the newcomer is phase-aligned with the existing mean; and the order parameter of the original M, computed on their phases alone, is unchanged to first order in K/M.

Proof. The bound is the triangle inequality applied to eq. (16): $|(M+1)\mathcal{R}_{\text{new}}| = |\sum_{j\le M} e^{i\phi_j} + e^{i\phi_{M+1}}| \le M\mathcal{R}_{\text{old}}+1$, with equality iff the added unit vector is parallel to the existing resultant. For the second statement, the coupling term in eq. (17) felt by an original member from the newcomer is $O(K/M)$, so the original members' mutual phase relations, and hence their restricted order parameter, are perturbed only at that order. □

Remark 7.15 (Why theorem 7.14 makes $\mathcal{R}$ diagnostic). Without it, a fall in $\mathcal{R}$ is ambiguous between "a new member joined whose rate differs" and "the existing coordination is decaying". Theorem 7.14 separates these: compute $\mathcal{R}$ on the original members alone. If it is unchanged, the drop is attributable to the newcomer's frequency mismatch and the panel is healthy; if it has fallen, the coordination itself is degrading. An empirical instance of the first case: a coordinated system at $\mathcal{R} = 0.99$ admits an additional strongly-mismatched component and the global figure falls to ≈ 0.91, while the same statistic restricted to the original components remains at 0.99.

7.5 Search, not fetch

Definition 7.16 (Search discipline)**.** A panel obeys the *search discipline* if every reasoner, on each query, reconstructs the relevant fragment of the network from Σ and the query, and no reasoner reads a stored answer keyed by a previous query.

Proposition 7.17 (Fetch is incoherent with theorem 4.10)**.** *Suppose a reasoner answers a query about node v by returning a cached value computed under a process Π', while the current process is $\Pi \neq \Pi'$. Then its answer is $\mathcal{T}(v,\{v\},\Pi')$, which by theorem 4.10 is the graph of a different circuit. No consistency condition relates it to the current one.*

Proof. Immediate from eq. (13): the value returned is the propagation at the cached process, and $\mathcal{T}$ is a function of Π among its arguments. □

Together with theorem 6.1, theorem 7.17 is why the search discipline is not a stylistic preference: caching answers in a system whose answers are process-indexed silently mixes graphs of different circuits.

7.6 Count-dependence is correct

Let m_i denote the number of acts a reasoner has performed, monotone non-decreasing and never reset. Probing the panel about node v at $m = 0$ and again at $m = 500$ should be expected to return different answers.

Proposition 7.18 (Stability under repeated probing is self-defeating)**.** *Suppose a panel is required to return an identical answer about v at all counts. Then either (a) the intervening acts had no effect on any reasoner's state, contradicting that acts are costly and state-changing; or (b) the panel computes its answer at count m by reasoning over its answer at count $m-1$, which is a different computation from the one that produced the earlier answer.*

Proof. The answer at count m is a function of the panel state at count m. If states at $m = 0$ and $m = 500$ differ, the answers differ unless the function is constant on the differing states, which is (a) in disguise for a function that actually depends on state. If the states do not differ, the 500 intervening acts changed nothing, which is (a). The only remaining way to force equality is to make the later computation take the earlier answer as input, which is (b). □

Remark 7.19. Theorem 7.18 removes cross-count answer stability from the list of things to validate. It is not a property the construction should have. What *is* validated is that m is monotone and never reset, and that the panel's membership boundary is not severed — the two conditions of section 8.

8 Identity of a panel

The panel is a persistent object that gets probed repeatedly, so it needs an identity criterion. Two candidates fail and one works.

Definition 8.1 (Boundary weights)**.** For reasoners A_i, A_j in a panel, the *boundary weight* $\partial(A_i, A_j)$ is the total weight of contacts between Γ_i and Γ_j in the panel's contact graph.

Proposition 8.2 (Identity is not internally readable)**.** *Let $\chi(A) = \min\{\rho(Q) : Q \text{ a cut of } \Gamma_A \text{ of rank} \geq 2\}$ be the panel's structural invariant. Under the search discipline (theorem 7.16), a reasoner reaches at most one cut per query and therefore cannot evaluate a minimum over all cuts. Hence χ is computable from outside the panel and not from inside it.*

Proof. χ is a minimum over the set of admissible cuts. A single query reaches one cut by theorem 7.16. Evaluating a minimum over a set requires access to the set; n queries reach at most n cuts and give an upper bound on χ, never the minimum, unless the cut set is exhausted — which is precisely the enumeration the search discipline forbids in a single query. □

Theorem 8.2 says the invariant that identifies the panel is perceived by its consumers, not by the panel. This is the right locus: an identity is what the outside reliably observes, and it does not decay on an internal clock.

Definition 8.3 (Continuity criterion)**.** A panel retains identity across a change if (i) χ is invariant under the change, and (ii) the act count m is not reset and no boundary weight $\partial(A_i, A_j)$ is severed to zero.

Proposition 8.4 (Replacement versus reconstruction)**.** *Replacing reasoners one at a time, each replacement preserving χ and leaving m and all boundaries intact, preserves identity. Dismantling the panel completely and rebuilding it to the same specification does not: the rebuilt panel has $m = 0$ and its boundaries were severed, failing theorem 8.3(ii) even though χ agrees.*

Proof. The first case satisfies both clauses of theorem 8.3 by hypothesis. In the second, dismantling severs every $\partial(A_i, A_j)$ and the rebuilt panel begins at $m = 0$, so clause (ii) fails. Clause (i) holding is not sufficient, since χ is invariant under relabelling and therefore cannot distinguish an original from a specification-identical copy. □

Remark 8.5 (Consequences that are not optional). Theorem 8.4 has three operational readings. Restarting a reasoner mid-panel yields a new individual, so its prior act count may not be carried over. Migrating a panel between hosts preserves identity exactly when m and the boundary structure are carried with it. Cloning a panel for throughput always produces a distinct individual, whatever the clone's χ, and its answers may not be attributed to the original.

Part V
Assessment

9 Divergence between derivations

Theorems 4.4 and 4.5 have a consequence for a practical situation that arises constantly: two routes to the same query return different counts.

Definition 9.1 (Amalgamation)**.** Given a query and a route r resolving a subset $\mathcal{U}_r$ of the network's contacts, the *amalgamation size* of r is $|\mathcal{U}_r|$ and its *unresolved remainder* is $\mathcal{E} \setminus \mathcal{U}_r$.

Theorem 9.2 (Divergence measures extent, not truth)**.** *Let routes r_1, r_2 answer the same query, tracking the same item and converging to the same fixed point x^*. Then their outputs cannot be in contradiction; their difference equals the difference of their amalgamation sizes, and the unresolved remainder of each is bounded below by β per unresolved contact.*

Proof. By theorem 4.5 the interior of each route is not part of the answer, so the routes cannot disagree about interiors. By theorem 4.4 inadmissibility arises only from non-convergence; both routes converge to x^* by hypothesis, so neither excludes any vertex the other admits on grounds of misalignment. What remains to differ is which contacts each route actually resolved, which is $|\mathcal{U}_{r_1}|$ against $|\mathcal{U}_{r_2}|$ by theorem 9.1. Under theorem 4.2 each unresolved contact contributes at least β to the remainder. $\square$

Corollary 9.3 (Reclassification)**.** *A reported inconsistency between two derivations that share endpoints and fixed point is not evidence that one is wrong. It is a measurement of how much of the network each derivation reached.*

Theorem 9.3 converts a class of findings from defects into measurements. It is conditional on a hypothesis that must be checked and not assumed: that both routes genuinely converge to the same x^*. Where that fails — where the two routes are answering different questions — the corollary does not apply and an inconsistency is exactly what it appears to be. A worked case, including the check and its outcome, is treated separately [Sachikonye, 2026].

10 Validation design

We state what a validation of this construction must contain, and what it must not.

(V1) **Circuit consistency.** The current law holds at every internal node of each test network to solver tolerance, and the Wegscheider condition holds on every independent cycle. Failure here invalidates everything downstream.

(V2) **Relabelling invariance.** The contact map is unchanged under permutation of species indices. This tests theorem 2.10.

(V3) **Coordinate irreducibility.** No axis of $\mathbf{S}$ is redundant *by construction*: reducibility on some network is a fact about that network and is to be reported, whereas reducibility on every network would show the third axis carries nothing. Triple coherence $\mathcal{R}$ is reported rather than thresholded.

(V4) **Address arithmetic.** Step-to-root equals address length at every depth tested, with zero variance. This tests theorem 3.7 and is a consistency check on the implementation, not evidence for a bound.

(V5) **Virtual fraction.** The fraction of sampled decompositions that are virtual is reported at several dispersion settings, together with recovery of the mean. This supports the mechanism of theorem 5.2, not its asymptotics.

(V6) **Panel floor.** The minimum separating family is obtained by exhaustive search over subsets of the available observables per theorem 7.3, and both directions of theorem 7.2 are checked on every subset. The floor is reported as computed, never assumed (theorem 7.4). Every pair of reasoners additionally carries a disjointness certificate per theorem 7.8, *including* the isomorphism check.

(V7) **Water-filling residual.** The solved allocation satisfies eq. (15) to a stated numerical residual, and the dropout boundary $\gamma_i'(0) \le p^\star$ is exhibited for at least one reasoner.

(V8) **Locking threshold.** K_c is computed from the observed frequency spread via eq. (18) and $\mathcal{R}$ is swept across it, rather than assumed.

(V9) **Localisation.** On admitting a mismatched reasoner, the restricted order parameter of the original members is reported alongside the global one, per theorem 7.14.

(V10) **Cache-freeness.** The full query batch is run with a materialised cache and with it deleted; outputs are compared byte for byte (theorem 6.3).

Explicitly *not* on the list: stability of answers across act counts, for the reason given in theorem 7.18.

Remark 10.1 (Controls must be able to fail). A control that cannot fail is worth nothing, and this is not hypothetical. In a related setting, a nearest-neighbour task over sequence fragments at $\approx 94\%$ identity was proposed as a control for whether a coordinate assignment carried physical content. The real assignment scored 1.00. So did a randomly shuffled assignment, with empirical $p = 1.000$: at that identity level the nearest neighbour is fixed by string identity alone, so *any* injective map ranks the near-identical pair first. The task could not discriminate physics from bookkeeping and was replaced by a graded task, which separated cleanly ($p < 0.005$). The episode belongs in the record rather than in an appendix, because a reader cannot distinguish a passing control from a non-discriminating one by looking at a table of successes.

The direct application here is (V6): a disjointness check that passes because the reasoners were constructed disjoint by fiat tests nothing. The isomorphism clause of theorem 7.8 exists to give the check a way to fail.

10.1 Results

The design above was executed as seventeen experiments over four networks — glycolysis, the citric-acid cycle, oxidative phosphorylation, and an EGFR/MAPK cascade — with thermodynamic parameters from standard compilations, k_{cat} values from enzyme kinetics databases, and concentrations from physiological reference ranges. Every stochastic experiment is seeded. Figures 1 to 6 report the outcome.

The consistency results hold as stated. Wegscheider residuals sit at machine epsilon on every enumerated cycle, which is the check that cannot be fudged: were it to fail, μ would not be a function on vertices and the circuit representation would be inadmissible. Relabelling leaves the contact map fixed to 10^{-12} while the control that permutes concentrations rather than labels moves it by six orders of magnitude more, so the invariance of theorem 2.10 is being tested rather than assumed. Step count equals address length with zero variance at every depth from 2 to 10;

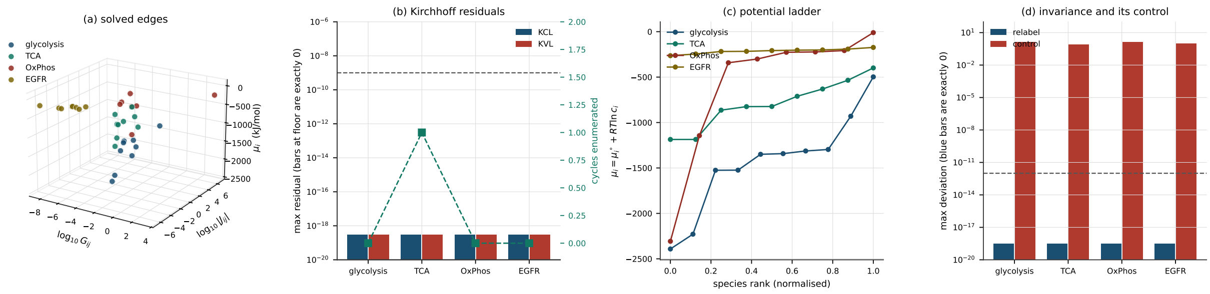

Figure 1: **The circuit layer.** (a) Every solved edge of all four networks in $(\log_{10} G_{ij}, \log_{10} |J_{ij}|, \mu_i)$: conductances span fourteen decades, so the circuit is stiff rather than uniform. (b) Kirchhoff residuals per network on a logarithmic axis; all bars are drawn at the floor because every residual is exactly zero. The right-hand axis gives the number of simple directed cycles enumerated, and it is the honest limitation of this check: only the citric-acid cycle contributes one, so the Wegscheider half of theorem 2.5 rests on a single cycle. (c) The potential ladder $\mu_i = \mu_i^\circ + RT \ln c_i$ against normalised species rank. (d) The invariance of theorem 2.10 against its control: relabelling moves the contact map by nothing at all, permuting the underlying concentrations moves it by roughly 10^0.

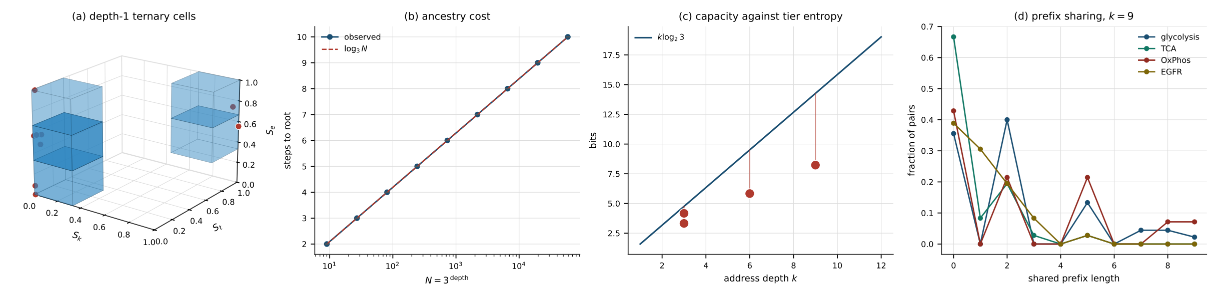

Figure 2: **Addressing.** (a) The depth-one ternary cells actually occupied by the four networks, drawn in the unit cube with opacity proportional to occupancy: the addressing scheme partitions $[0, 1]^3$, but the coordinates populate a small part of it. (b) Ancestry cost. Observed steps to root against $N = 3^{\text{depth}}$ on a logarithmic axis, lying on $\log_3 N$ with zero variance at every depth — the content of theorem 3.7(ii), computed without materialising the trie. (c) Capacity $k \log_2 3$ against address depth, with the four tiers of the worked family plotted at their entropies; every tier sits below the line, which is the inequality of theorem 3.13 holding with slack. (d) Distribution of shared prefix length over all species pairs at $k = 9$. Mass concentrated at short prefixes is what makes the address discriminating.

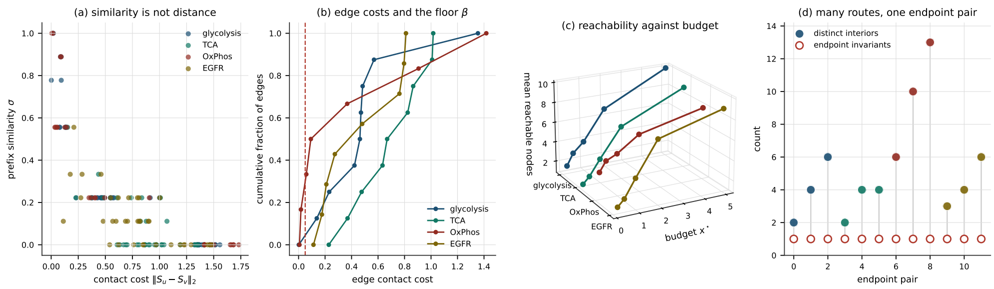

Figure 3: **Propagation.** (a) Prefix similarity σ against contact cost $\|S_u - S_v\|_2$ for every species pair. The relation is not monotone — pairs at $\sigma = 0$ occur at every cost, and pairs at equal cost take most available σ — which is theorem 3.11 shown directly: σ reports a partition, not a distance. (b) Cumulative distribution of edge contact costs, with the floor β marked; the mass of edges above β is what makes budgets bind. (c) Mean reachable nodes against budget $x^\star$ per network. Reachability grows with budget at a network-specific rate, so the budget is a real constraint rather than a formality. (d) For each endpoint pair, the number of distinct route interiors against the number of distinct endpoint invariants. Interiors proliferate; the invariant stays at one. This is theorem 4.5: the construction returns an equivalence class of routes, and increasing k does not narrow it.

the equivalence of theorem 3.10 holds with no mismatches; deleting the cache changes no output byte.

Two results are negative, and they are the informative ones.

The coordinate degenerates under a dominating species. On oxidative phosphorylation, S_k and S_t are exactly collinear: each regresses on the other two axes at $R^2 = 1.000$. The cause is visible in the inputs. Water at $55\,\mathrm{mol\,L^{-1}}$ exceeds the next-largest concentration by roughly six orders of magnitude, so both max-normalisers in eq. (5) collapse onto a two-valued indicator separating the water/ATP pair from everything else, and the address carries about one axis of information instead of three. Glycolysis sits close to the same edge at $R^2 = 0.989$. This is a defect of the normalisation, not of the networks: eq. (5) divides by network extrema and is therefore not robust to a species that dominates the extremum. On the remaining three networks no axis is recoverable from the other two, so the third axis is not redundant by construction — which is the claim (V3) actually makes — but a practitioner should compute the axis regressions before trusting a three-axis address on any given network.

The panel floor is two, not three. Theorem 7.2 holds exactly: across all subsets of the available observables, separating every pair and separating the whole state set coincide with no exceptions in either direction. The floor asserted by the earlier reading of theorem 7.3 does not survive. Exhaustive search returns $\chi = 2$, attained uniquely by absorption together with oxidation state, for the reason given in theorem 7.4. We report this rather than adjusting the state table to recover the number three, since a state table tuned until the answer comes out right is the failure mode theorem 10.1 describes. What survives is the substantive claim, and it is the one worth carrying: disjointness of observables buys resolution, cardinality does not, and the floor must be computed per system.

One further correction belongs in the record. The discrimination experiment initially modelled each probe as a bin label — band position rounded to the instrument's resolving power — and reported that absorption alone separates all seven states. That was an artefact: rounding separates two readings whenever they straddle a bin edge, however close they are, so 417 nm and 418 nm came apart while genuinely-resolved pairs did not. A resolving power is a proximity relation, not a partition. Rewriting each probe as the separation relation it actually denotes produced exactly

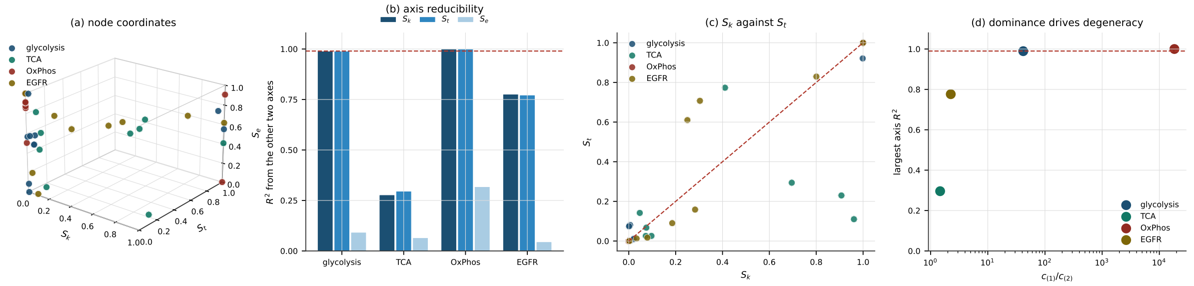

Figure 4: **The coordinate, and where it degenerates.** (a) Every species of all four networks at its coordinate $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ in the unit cube of eq. (5). (b) Axis reducibility: R^2 from regressing each axis on the other two, with the 0.99 line marked. Oxidative phosphorylation saturates on S_k and S_t and glycolysis nearly does; the citric-acid cycle and the EGFR cascade do not. (c) The oxidative-phosphorylation collinearity shown without a regression: S_t against S_k collapses onto a two-valued indicator rather than spreading over the square. (d) The mechanism. Largest axis R^2 against the concentration dominance ratio $c_{(1)}/c_{(2)}$ on a logarithmic axis. Degeneracy tracks dominance: the two networks carrying a species that exceeds the next by orders of magnitude are the two whose coordinate collapses. Since eq. (5) divides by network extrema, this is a defect of the normalisation and not of the networks.

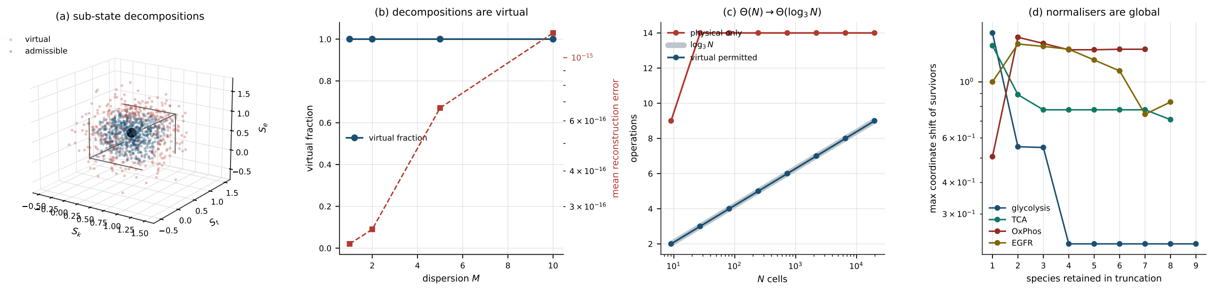

Figure 5: **Virtual sub-states and the collapse.** (a) One state (black) and its sub-state decompositions. Sub-states falling outside the unit cube (red) are virtual in the sense of theorem 5.2 — they are not themselves admissible coordinates, yet they average exactly to one that is. (b) Virtual fraction against dispersion M, with mean reconstruction error on the right-hand axis. Every decomposition at every dispersion is virtual, and reconstruction holds to 10^{-15}: the averaging identity is exact while the parts are unphysical. (c) The collapse of theorem 5.3. Operations against N cells, logarithmic in N; permitting virtual sub-states tracks $\log_3 N$ (drawn as the wide underlay, which the data lies on) where the physical-only count saturates. (d) Why the normalisers are global. Truncating a network to its largest species shifts the coordinates of the survivors, by up to order unity for small truncations — so a coordinate is a statement about a network, not about a species in isolation.

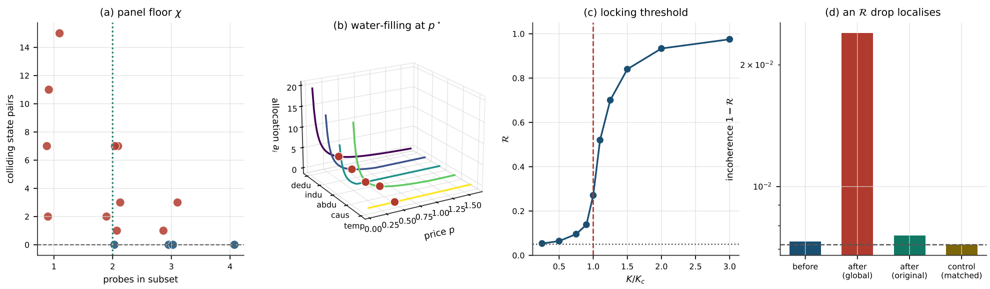

Figure 6: **The panel layer.** (a) Colliding state pairs for every subset of the four observables, against subset size. Subsets reaching zero collisions (blue) are exactly those that separate the whole state set, with no exceptions in either direction, which is theorem 7.2. The floor $\chi = 2$ is marked; size alone does not determine separation, since some three-element subsets still collide. (b) Water-filling. Allocation against price for each reasoner, with the solution at the single shadow price $p^\star$ marked. Allocations are graded and the low-weight, high-cost reasoner sits at the dropout boundary of theorem 7.10 rather than being excluded by a separate rule. (c) The locking threshold of theorem 7.13. Order parameter against K/K_c with K_c computed rather than fitted, and the finite-M incoherent floor $1/\sqrt{M}$ marked; the transition occurs at the predicted place. (d) Localisation, plotted as incoherence $1 - \mathcal{R}$ on a logarithmic axis because the whole effect lies in the third decimal. Admitting a mismatched newcomer drops the global order parameter, the order parameter restricted to the original members is essentially unmoved, and the matched-newcomer control does not drop at all. The dashed line is the bound of theorem 7.14.

the two documented collisions.

11 Limitations

Steady state is assumed throughout. Theorem 2.5 and everything after it hold at steady state. The recovered graph is time-invariant, and no claim is made about transient dynamics, oscillatory regimes, or systems far from stationarity. A network whose biology is in its transients is outside scope.

Ideal-solution chemical potentials. Equation (1) omits activity coefficients. For crowded intracellular conditions this is a real approximation, and the coordinate of eq. (5) inherits its error. The error enters S_e directly and $S_\mathrm{k}, S_\mathrm{t}$ through the flux and conductance sums.

Mechanism is not recoverable. By theorem 4.5 the construction returns an equivalence class of routes. Where the route is the scientific object — mechanistic enzymology, isotope tracing — this is the wrong instrument, and increasing k does not help.

Coordinates are network-relative. The normalisers in eq. (5) are network extrema, so addresses are not comparable across networks without a shared normalisation. Comparing addresses computed under different networks is the most likely misuse.

The normalisation is not robust to a dominating species. Dividing by a network extremum makes the coordinate fragile when one species carries that extremum by orders of magnitude. Section 10.1 exhibits the failure: on oxidative phosphorylation, water at bulk concentration collapses S_k and S_t into the same two-valued indicator, giving exact collinearity and reducing an address that nominally carries three axes to about one. A rank or logarithmic normalisation would likely be better behaved, but we have not established that and do not use it, so the practical guidance is to compute the axis regressions of (V3) per network and to distrust the address where an axis is recoverable from the others.

Kinetic parameters bound accuracy. Theorem 2.10 says the graph is a function of the kinetic description. Where k_{ij} or c_i are wrong, the graph is wrong in a way no internal check detects, because internal consistency is preserved under consistent error.

Two constants are unreconciled. Whether the contact floor β of theorem 4.2 and the numerical resolution floor of the coordinate construction are the same constant under rescaling is open. We have not established it and do not use it.

Metric compatibility is unverified. The Euclidean contact cost of eq. (6) and the information-geometric metric natural on a space of normalised triples [Amari, 2016] need not agree. We use Euclidean distance because the reference implementation does, and flag that no compatibility result is claimed.

The panel results are structural, not empirical. Theorems 7.9, 7.13 and 7.14 are theorems about the panel model. That a panel of reasoners built this way outperforms a single reasoner on a biological question is not established here and would require the validation of section 10 run end to end on a real network.

12 Conclusion

The construction reduces to three identifications and one accounting.

The identifications: a steady-state reaction network *is* a resistive circuit (theorem 2.5); the solved circuit's per-species coordinate *is* a base-3 address under interleaved digitisation (theorem 3.2), so that ancestry and similarity are both prefix arithmetic (theorem 3.7, theorem 3.10); and the causal knowledge graph *is* the circuit's propagation operator evaluated at the resting process (theorem 4.10). Nothing is asserted into the graph; it is recovered, and recomputing returns it.

The accounting: backward interrogation costs $\Theta(\log_3 N)$ and forward construction costs $\Theta(N)$, with the gap attributable entirely to virtual sub-states, which are legitimate in the first case and provably unavailable in the second (theorem 5.2, theorem 5.3, theorem 5.4). Resident state is $O(1)$ in addressable objects, bought by recomputation rather than compression, and worth having only where the query set is not known in advance (theorem 3.4, theorem 3.5).

Two consequences are worth separating from the machinery because they change how results are read. First, an edge of the recovered graph names a class of mechanisms and not a mechanism (theorem 4.5); this is a limitation, and stating it as a theorem is the only honest way to carry it. Second, two derivations that share endpoints and fixed point cannot contradict each other, so their divergence measures resolved extent rather than truth (theorem 9.2, theorem 9.3) — conditional on a convergence check that must actually be performed.

What the construction does not supply is accuracy. Theorem 2.10 guarantees the graph is a faithful function of the kinetic description it was given, and says nothing whatever about whether that description is right.

The validation of section 10.1 bears this out in both directions. The consistency claims hold to machine precision, and two claims that were stated too strongly do not: the coordinate degenerates on a network containing a species at bulk concentration, and the panel floor is a computed property of the state and observable sets rather than the constant three. Both were found by tests built to be capable of failing, which is the only reason they were found at all.

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